THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. In a number pyramid each block is the sum of the two below it. The bottom row is 1, 2, 3, 4. The top block is: 1

(a) 10 **(b)** 15 **(c)** 22 **(d)** 20

2. In a quadrilateral, a diagonal of 20 cm has perpendiculars of 6 cm and 4 cm drawn to it from the opposite vertices. Its area is: 1

(a) 30 cm² **(b)** 100 cm² **(c)** 60 cm² **(d)** 200 cm²

3. The sum of the consecutive triangular numbers T_6 and T_7 is: 1

(a) 36 **(b)** 56 **(c)** 49 **(d)** 50

4. Which of the following equals 4.07×10^{-3}? 1

(a) 4070 **(b)** 0.00407 **(c)** 0.0407 **(d)** 0.000407

5. A Babylonian numeral has 2 group(s) of 60 and 15 units. Its value is: 1

(a) 17 **(b)** 105 **(c)** 135 **(d)** 27

6. The diagonals of a rectangle ABCD meet at O and $\angle AOB = 80^\circ$. Then $\angle OAB$ is: 1

(a) 100° **(b)** 40° **(c)** 80° **(d)** 50°

7. The sum of the three consecutive numbers 12, 13 and 14 is: 1

(a) 42 **(b)** 36 **(c)** 39 **(d)** 37

8. $(x + 2)(x + 5)$ equals: 1

(a) $x^2 + 7x + 7$ **(b)** $x^2 + 10$ **(c)** $x^2 + 10x + 7$ **(d)** $x^2 + 7x + 10$

9. 8 workers can finish a job in 15 days. 12 workers will finish it in: 1

(a) 10 days **(b)** 11 days **(c)** 22.5 days **(d)** 12 days

10. The marked price of a jacket is ₹ 2500 and it is sold for ₹ 2000. The discount percent is: 1

(a) 40% **(b)** 20% **(c)** 22% **(d)** 18%

11. A triangle has sides 14 cm, 48 cm and 50 cm. It is: 1

(a) cannot be said **(b)** not right-angled **(c)** equilateral **(d)** right-angled

12. In a survey of 60 students, 14 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 168 **(b)** 84 **(c)** 42 **(d)** 94

13. A cube of side 3 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 8 **(b)** 1 **(c)** 6 **(d)** 12

14. A plant's height (cm) at the end of weeks 1 to 6 is 4, 8, 15, 19, 22, 28. Between which two consecutive weeks was the growth the greatest? 1

(a) Between week 3 and week 4 **(b)** Between week 2 and week 3

(c) Between week 4 and week 5 **(d)** Between week 1 and week 2

15. In a calendar the nine dates in a 3×3 block have the middle date 10. Their sum is: 1

(a) 81 **(b)** 30 **(c)** 90 **(d)** 99

16. The cost of flooring a 18 m by 12 m hall at ₹ 45 per square metre is (in ₹): 1

(a) 1350 **(b)** 2700 **(c)** 9720 **(d)** 261

17. For m = 4, the Pythagorean triplet $(2m,\ m^2-1,\ m^2+1)$ is: 1

(a) 8, 15, 17 **(b)** 8, 17, 18 **(c)** 4, 15, 17 **(d)** 8, 14, 18

18. $\frac{1}{64}$ expressed as a power of 2 is: 1

(a) 2^{-7} **(b)** 2^{6} **(c)** 2^{-5} **(d)** 2^{-6}

19. A Maya numeral has 3 in the 20s place and 12 in the units place. Its value is: 1

(a) 15 **(b)** 48 **(c)** 42 **(d)** 72

20. In a kite ABCD with AB = AD and CB = CD, $\angle B = \angle D = 90^\circ$ and $\angle A = 70^\circ$. Then $\angle C$ is: 1

(a) 200° **(b)** 110° **(c)** 90° **(d)** 130°

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Show that the sum of 16 and its reverse is divisible by 11. 2

22. The area of a rectangular field is 1200 m² and its length is 48 m. Find its breadth and perimeter. 2

23. Find the Pythagorean triplet whose smallest member is 6. 2

24. Write $\left(\frac{2}{3}\right)^{-3}$ in simplest form. 2

25. Write 29 in the binary system. 2

26. In a parallelogram ABCD, $\angle A = (3x+10)^\circ$ and $\angle B = (2x-10)^\circ$. Find x and $\angle A$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Find XLIV + LXVII and write the answer in Roman numerals. 3

28. ABCD is a rhombus whose diagonals meet at O. If $\angle BAD = 64^\circ$, find $\angle OAB$, $\angle ABC$ and $\angle OBA$. 3

29. Apply the Kaprekar routine to 3521 repeatedly until you reach 6174. Write each step. 3

30. Find 53×47 using the identity $(a+b)(a-b) = a^2 - b^2$. 3

31. 10 men can build a wall in 18 days. How many days will 15 men take to build it? 3

OR

Find the amount and the compound interest on ₹ 12500 for 2 years at 8% per annum compounded annually. 3

32. A ladder 29 m long reaches a window 21 m above the ground. Find the distance of the foot of the ladder from the wall. 3

OR

A family's monthly expenditure: food 30%, rent 25%, education 20%, others 25%. Find the central angle of each sector of a pie chart. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Take 83 and its reverse 38. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

34. Prove that $(a+b)^2 - (a-b)^2 = 4ab$ and verify it for $a = 7$, $b = 3$. 5

35. A bus covers 240 km in 4 hours at a uniform speed. Find 5
(i) its speed
(ii) the distance it covers in 7 hours
(iii) the time it takes to cover 360 km.

36. A refrigerator has marked price ₹ 30000. The shopkeeper allows a discount of 10% and then charges GST of 18% on the discounted price. Find the amount payable. 5

OR

Two poles of heights 20 m and 12 m stand on level ground 15 m apart. Find the distance between their tops. 5

37. The marks of a student are: Maths 18, Science 15, English 9, Hindi 12, SST 6. Total 60. Find the central angle of every sector of a pie chart and the percentage of the total taken by Maths. 5

OR

A cube of side 9 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 729. 5

38. The runs of two players in 5 matches are: A: 80, 18, 34, 43, 30; B: 39, 49, 41, 20, 36. Find the mean of each and say who is more consistent (smaller range). 5

OR

Think of a number n. Double it, add 10, halve the result and subtract n. Show algebraically that the final answer is always 5. Verify for $n = 7$. 5

END OF QUESTION PAPER **Total: 80 Marks**